

# Vandad Davoodnia

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## Summary

I am a Ph.D. candidate at Queen's University, specializing in computer vision, machine learning, and generative models with a strong focus on human-centric applications. With 8+ years of research experience in academia and 1.5+ years in industry (internship at Ubisoft Toronto), I look forward to new challenges and opportunities. I am passionate about designing, developing, and deploying large-scale AI solutions that enhance virtual worlds, simulations, and human interactions.

## Experience



### Research Assistant - Ph.D. Candidate

Queen's University

Sep 2018 - Present (5 years 9 months)

- Explored solutions for pose estimation from ambiguous pressure maps using self-supervision, GANs, and domain adaptation, resulting in five publications. (Publications, Pytorch, Python)
- Was awarded 30,000 CAD\$ through Mitacs accelerated internship program to support 8 month of research on deep learning solutions for markerless MoCap at Ubisoft Toronto.
- Assisted in publishing four papers on supervised and semi-supervised EEG learning, enhancing user experience in a car using e-textiles, and cancer prediction. (Publications, Pytorch, Python)
- Explored differentiable rendering, multi-view geometry, and self-supervision for pose estimation. (Pytorch, C++, Cuda)
- Managed lab's GPU servers, honing communication, hardware, administration, and security skills.
- Completed courses in NLP, Machine Learning, Deep Learning, Artificial Intelligence
- Created an NLP model for detecting irony on Tweets, achieving 10% improvement over the top competitor (Tensorflow, Python)
- Developed an ML solution for predicting heavily loaded lifts from EMG signals with less than 100g error (Tensorflow, Python)
- Developed a hand movement tracking software using 3DsMax, Vicon MoCap, and Unity 3D



### Research And Development Intern

Ubisoft

May 2022 - Jan 2024 (1 year 9 months)

- Designed, developed, and deployed a markerless MoCap app for outdoor recording using GoPro and iPhone cameras to enable low-cost and portable prototyping. (Docker, Gradio, FastAPI, and Kubernetes)
- Designed a markerless MoCap solution achieving competitive results with top industry competitors and top performance among academic models by 47% error reduction on out-of-distribution data. (Publications)
- Published two papers on markerless MoCap methods. (Publications, PyTorch, MMEngine)
- Investigated synthetic data generation and developed a parametric 3D body model with blend shapes. (PyTorch, Blender, MoBu, Python)

- Explored generative models, including VAE, VQ-VAE, GAN, and diffusion for motion generation.

## **Reasercher Assistant**

Brain Engineering Center, Institute for Research in Fundamental Sciences, IPM

Sep 2015 - Jul 2018 (2 years 11 months)

- Developed Eye Speed app for diagnosing glaucoma patients. (Publication, iOS/Objective-C, Android/Java)
- Developed an ANFIS model trained by PSO to predict biomass behavior. (Publication, MATLAB, Evolutionary Algorithms, Swarm Optimization)
- Developed a novel response tuning method by analyzing neuron activity time series. (Publication)
- Published 3 SFN abstracts in collaboration with Dr. Gottlieb (Columbia University) and Dr. Alonso (SUNY).
- Collaborated with Blackrock Microsystems Inc. to develop a an add-in for the NPlayServer app, enabling synchronized video and multi-unit electrophysiological recordings. (C++, Qt, OpenCV)
- Assisted in setting up an electrophysiological lab and implemented a real-time experimental control system for primate behavioral studies. (MATLAB, Windows SDK, C)
- Performed MRI segmentation to inform surgical decision-making. (MATLAB)
- Analyzed the relation between SUA and LFP properties in primate visual cortex, honing signal processing and statistical analysis skills. (MATLAB)

## **Software Developer**

Sep 2012 - Jun 2018 (5 years 10 months)

- Developed an extension in Transcad/GISDK for simulating Iran transit road and rails to facilitate users' experience
- Developed a billiard game (C, SDL)
- Developed a cellular automata sandbox (C++)
- Developed scientific calculators (C++/Arduino, Java)
- Developed 20+ games and simulation programs (Python, C, C++, Objective-C, Java, MATLAB)

## **Education**



### **Queen's University**

Doctor of Philosophy - PhD, Electrical and Computer Engineering

Sep 2018 - May 2024



### **Sharif University of Technology**

Bachelor's Degree, Electrical and Electronics Engineering

2012 - 2017

Thesis on 2D Micro-electrode-based Active Extra Cellular Recording System

## **Salam**

High School, Mathematics

2008 - 2012

Ranked 4th in Nationwide University Entrance Exam (Concours)

## **Skills**

FastAPI • Gradio • Streamlit • Docker • Machine Learning • Qt • Kinect SDK • Objective-C •  
Arduino • PyTorch

## **Honors & Awards**

**Ranked 4th in Nationwide Iranian University Entrance Exam (known as  
Concours) for B.Sc. Admission Among Over 500,000 High School Graduates  
Taking the Exam**

Jun 2012

**Passed the First Round of Iranian National Chemistry Olympiad**

Feb 2011

**Passed the First Round of Iranian National Computer Olympiad**

Feb 2011

**Passed the First Round of Iranian National Mathematics Olympiad**

Feb 2011

**Passed the First Round of Iranian National Astronomy Olympiad**

Feb 2011

**Passed the First Round of Iranian Computer Skills Competition**

Feb 2010